

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. We separate three conditions: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared Lipschitz encoders, and the bound grows as the square root of the population read, so on a large population it constrains nothing. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor; a broad family needs input-conditioned maps and discriminating codes. Lean 4 verifies these implications, and one model satisfies all eight connecting hypotheses together; this shows

consistency, not cortical realization. The conditions yield a disconfirmable comparison: decoders fitted independently on overlapping regions should agree more on reported than on matched unreported stimuli at equal phase coherence. An exploratory EEG analysis is a measurement check only; its pooled estimator places a perfectly synchronized signal at zero coherence. Resource and implementation constraints, and a held-out onset comparison during anaesthetic emergence, are stated briefly. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012; Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields; identifying them with perceptual content requires independent decoding evidence.

Binding-by-synchrony and communication-through-coherence accounts

17 treat phase relations as the medium of integration or of selective routing
 18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
 19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
 20 leave open what fact about synchronized activity makes it one content rather
 21 than coordinated activity carrying several. The contribution here is to sep-
 22 arate the three requirements and prove which follow from which. Phase
 23 order constrains shared content only through encoders related on overlaps,
 24 with a bound that grows as the square root of the population it is read
 25 on. Compatible local descriptions, not synchrony, determine a unique global
 26 state. A contracting self-map fixes one state but can reconstruct accurately
 27 only a family of bounded diameter. Each result names a measurement that
 28 phase data alone cannot supply.

29 The familiar noisy mean-field oscillator supplies a stationary coherence
 30 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource
 31 condition for a declared actuator, bounds that connect phase order to con-
 32 tent only through specified encoders, and conditions under which compatible
 33 descriptions glue and an accurate reconstruction fixes the glued state. A
 34 Lean 4 development checks these implications and a joint witness satisfies
 35 their eight connecting hypotheses in a specified model. It proves neither that
 36 cortex satisfies those hypotheses nor that the glued state is experienced. The
 37 latter identification motivates the framework and needs its own endpoint.

38 The empirical question is which measurements could support or discon-
 39 firm the proposed connections. A phase resultant can miss a locally or-

40 dered winding field; stationary phase distributions can leave the coupling
 41 mechanism unidentified; and coherent adaptation can lose alignment with
 42 a designated content template. Independent decoders must identify shared
 43 quantities before their agreement can be tested. We therefore distinguish sta-
 44 tionary branches from finite observed onset, and compare prespecified field
 45 and connectivity onset models on held-out emergence recordings. The com-
 46 parison tests a specified spatial prediction, not the presence of experience.
 47 Section 10.2 states the comparison that bears on conscious content directly.
 48 Sections 2–5 develop the three requirements; Section 6 applies their mea-
 49 surement obligations to a digital candidate. Section 7 states the resource and
 50 composition premises, and Sections 8–9 examine observables and the cortical
 51 test. Table 1 lists the eight connecting hypotheses; Supplemental Material
 52 gives their derivations and controls.

53 **2 Limits of phase order**

54 Several limits hold without the cortical hypothesis. Equal phases can coexist
 55 with incompatible local descriptions; phase order constrains shared content
 56 only through encoders related on overlaps. A winding field can have zero
 57 global resultant while remaining locally ordered. A constant readout can
 58 have a unique fixed point without reconstructing its input. A region that
 59 does not distinguish relevant states cannot encode them, and a site cannot
 60 report a change outside its causal past by a declared deadline. These lim-

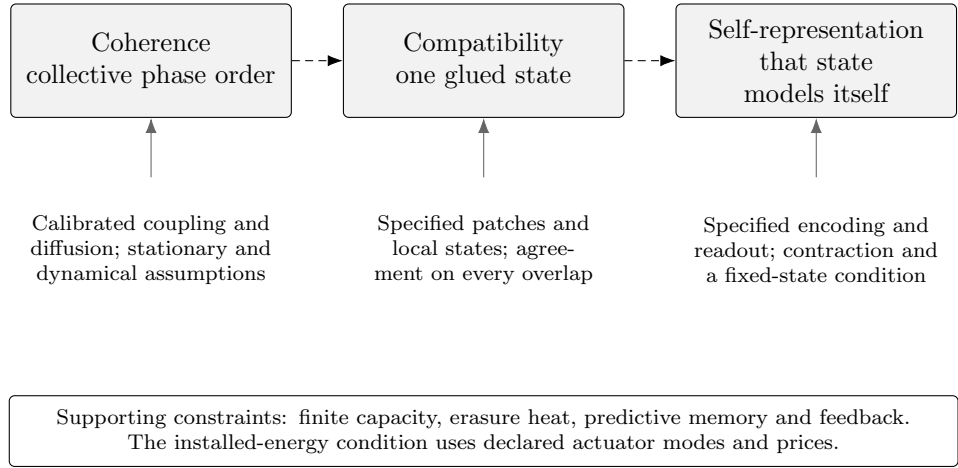


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

61 its reflect familiar phase oscillator and information constraints (Acebrón et
 62 al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017).
 63 The sections below state each one beside the condition or measurement it
 64 motivates.

65 3 Coherence: physical coupling and its re- 66 source condition

67 3.1 Dynamics and the stationary threshold

68 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

69 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
70 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
71 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
72 quency heterogeneity and diffusion are distinct parameters. The stationary
73 calculation below uses identical frequencies and this scalar coupling. Ap-
74 plying it to a spatial kernel requires an aggregation rule and a check of the
75 resulting approximation.

76 For identical frequencies in their rotating frame, the positive normalized
77 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

78 where I_n are modified Bessel functions and ψ is the collective phase. The for-
79 mal derivation uses the absence of stationary current for a periodic gradient
80 drift to obtain this density, then verifies normalization and self-consistency.

81 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
 82 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
 83 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

84 Figure 2 shows this stationary branch.

85 Linearization about the uniform density gives rate $K/2 - D$ for the first
 86 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 87 exactly above threshold. About the coherent density, the quadratic form
 88 is strictly dissipative transverse to collective rotation, whose rate is zero.
 89 These linear estimates concern the limiting equation. Relating them to finite
 90 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 91 tion; exchange symmetry alone does not establish the needed independence
 92 (Sznitman, 1991).

93 **3.2 Installed energy bounds mean coupling**

94 A spatial model replaces the finite sum by integration against a kernel. One
 95 declared implementation carries response modes with continuous profiles ϕ_i ,
 96 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_\Xi(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

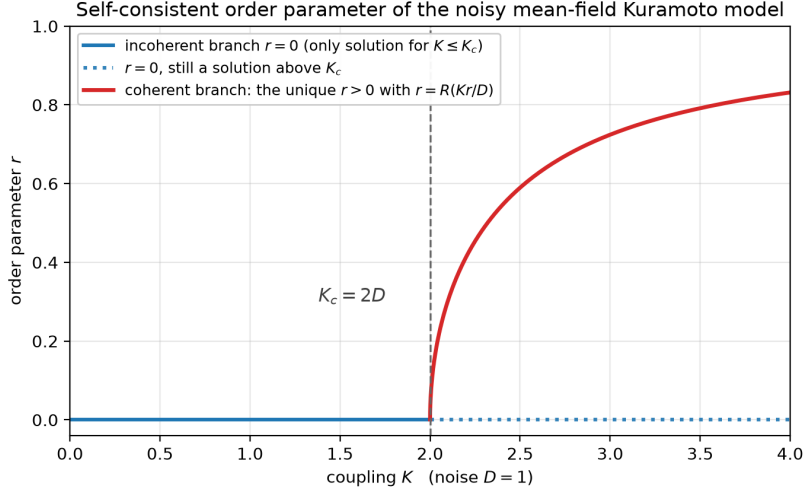


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

97 The same configuration sets the reciprocal kernel and stored energy. An
 98 executed transition changes the kernel through its changed occupancies and
 99 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 100 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 101 pair reconstruction on refining spatial partitions converges to its product-
 102 measure integral under the specified compactness and sampling assumptions;
 103 the refinement index is independent of the number of control updates.

104 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 105 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

106 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned} \tag{5}$$

107 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 108 coherent stationary branch and any growing Fourier mode about the uniform
 109 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 110 is necessary: energy above it need not produce sufficient coupling because
 111 the bound can be loose.

112 The theorem concerns stored energy in the declared actuator modes. The
 113 estimated cortical signalling expenditure and extracellular electrostatic en-
 114 ergy in Supplemental Material measure different quantities; neither identifies
 115 the occupancies, mode prices or their relation to metabolic supply. Conse-
 116 quently the bound supplies a necessary condition for a declared actuator,
 117 but no cortical value of κ or demonstration that cortex meets it. Applying
 118 the scalar threshold to heterogeneous or signed spatial dynamics also needs
 119 a justified reduction.

120 4 Compatibility: from coordinated activity 121 to a global description

122 4.1 Why common timing is insufficient

123 For overlapping patches U and V , compatibility means that restricting either
124 local description to $U \cap V$ gives the same object. Descriptions can agree on
125 shared quantities while contributing different information elsewhere. Com-
126 patible sections of the specified sheaf have a unique global extension. On
127 a finite cover, local finite spatial measures agreeing as measures on every
128 overlap also have a unique finite-measure extension whose sheaf image is the
129 glued section. No mass-one normalization is imposed.

130 In the cup example, visual and reaching descriptions must express a
131 shared quantity, such as cup location, in common coordinates. Restriction
132 retains that quantity while dropping details particular to each description.
133 Agreement permits surface appearance and hand configuration to contribute
134 to one situation. The mathematical counterexample assigns the same phase
135 to every patch while two local mass profiles disagree on a shared site. Those
136 profiles have no common extension. Because they are chosen independently
137 of phase, synchrony places no constraint on their disagreement.

138 This distinction limits the biological interpretation. Receptive territories
139 and content domains must be related to spatial patches, and the descriptions
140 must support the required extension. Matching overlap marginals of de-

141 coded probability laws need not determine a unique global joint distribution
 142 (Abramsky & Brandenburger, 2011).

143 The cover is itself a choice, and the construction constrains it in one way
 144 only: synchronized patches agree where they overlap. Where they do not
 145 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 146 satisfies that obligation for *every* assignment of local descriptions and phases
 147 and the gluing theorem still returns a unique global section, which is the
 148 tuple it was handed. Membership of the class then reports nothing about the
 149 system beyond the patches covering it. What a gluing claim asks of a cover
 150 is therefore a non-degeneracy the theorem does not assume: that distinct
 151 patches meet, and that what they share is a quantity fixed independently of
 152 the agreement it is being used to establish.

153 4.2 When coherence does constrain content

154 A declared relation between phase and content gives a positive result. Let
 155 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 156 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 157 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 158 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 159 sites; their equality on overlaps is a supplied relation between descriptions of
 160 the shared quantity. Write $z_k = e^{i\theta_k}$ and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The
 161 phasors' squared deviations from their mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$,
 162 and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$ gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair.

163 Consequently every overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1-r^2)}. \quad (6)$$

164 Here discrepancy is the norm of the difference, and r and N are read on
165 whichever population the bound is applied to. That choice is what decides
166 whether the estimate says anything. A patch carries an order parameter
167 of its own, and reading the bound there carries that patch's resultant and
168 that patch's site count, so a large population observed through small patches
169 is constrained by the small count. Read instead on the whole array, the
170 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
171 from an average over N sites. The order cannot be improved in general:
172 one oscillator in antiphase among N leaves $N(1-r^2)$ near 4. At $r = 1$ the
173 observations agree exactly at either scale. A shared scalar encoder is a special
174 case. Dropping the overlap relation permits disagreement even at perfect
175 locking. A common nonexpansive linear projection preserves agreement, but
176 agreement of a projection does not establish agreement of the full vector.

177 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
178 constrains anything only where the right-hand side falls below that. At
179 the sheet's own patch order, the bound is informative for at most 10 sites;
180 beyond that, the coherence-to-content link requires the bridge assumption
181 (E78) of the cortical hypothesis. Supplemental Material provides the detailed
182 derivation of this saturation, and states three conditional derivations that

183 replace the uniform factor $\sqrt{N}$ by a count of disagreeing pairs, a chain across
 184 a cover's shared sites, or a spectral gap at a frequency-locked configuration.

185 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 186 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 187 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 188 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

189 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 190 duce an initial mismatch. Approximate observations leave a nondecaying
 191 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 192 update to the phase process.

193 4.3 What approximate agreement determines

194 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 195 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 196 Weights that vanish outside their patches and sum to one at each site select
 197 a global profile by weighted averaging. The selection lies within ε of each
 198 patch there, and any two such selections differ by at most ε , independently
 199 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 200 $L\varepsilon$ across selections.

201 At zero tolerance, the selection is the unique exact extension. At every
 202 positive tolerance there are families with no exact extension; selection then

203 gives an approximation whose dependence on weights is bounded. Supple-
 204 mental Material gives a three-site witness attaining the bound. The mass-
 205 profile construction uses the scalar case of the content dynamics and addi-
 206 tionally requires nonnegativity of the initial profiles and observations. The
 207 general vector result supplies no identification with the spatial-measure sheaf.
 208 These constructions concern declared objects: a specified cover, contents
 209 valued in a normed space, and encoders related on overlaps. They select
 210 neither the neural cover nor the decoder, and identifying their contents with
 211 perceptual content, or establishing that the identification bears on experi-
 212 ence, is a further modelling task; Section 8 sets out what a measurement of
 213 it would have to establish.

214 **5 Self-representation: a state that models it-** 215 **self**

216 **5.1 The reflexive construction**

217 Compatibility gives a global state. Reflexivity relates it to an encoding within
 218 that state. Equip global sections with a complete metric. If the additional
 219 restriction-resonance condition identifies encoding with restriction to a region
 220 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

221 The region and readout are fixed across the state space. If F contracts and
 222 fixes the section s_* formed from the local descriptions, Banach's theorem
 223 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
 224 plies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Su-
 225 percritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$
 226 less than one for a relaxation time $\tau > 0$. Constructing a map with that
 227 factor remains a modelling obligation. A fixed contraction can reconstruct
 228 accurately only a restricted family of states: if its error is at most ε through-
 229 out that family, any two members are at most $2\varepsilon/(1 - \Lambda)$ apart.

230 In the cup example, the global state includes the represented scene and the
 231 perceiver's relation to it. A local encoding must retain enough information
 232 for the readout to recover that modelled state; cup position alone would be
 233 insufficient if hand configuration and viewpoint could not be reconstructed.

234 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 235 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

236 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 237 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 238 weakens.

239 5.2 Reconstruction across a family of states

240 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 241 encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that
 242 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

243 Thus M relevant states pairwise more than 2ε apart require at least M
 244 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 245 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 246 existing self-prediction map.

247 This is an additional accuracy criterion on a family, with no requirement
 248 for multiple exact fixed points. For the same fixed contracting map used by
 249 E89, however, the diameter bound above is an additional restriction: at zero
 250 error the family contains at most one metric-distinct state. A broad family
 251 therefore requires a different, possibly input-conditioned reconstruction map
 252 with its own accuracy and stability assumptions. A blind encoder or constant
 253 readout fails the stated criterion for separated states despite having a unique
 254 fixed point.

255 Where the encoding is read as the state's own restriction to a region, the
 256 criterion becomes a condition on the region. Restriction resonance requires
 257 the encoder and that restriction to identify the same pairs of states, so a
 258 boundary reconstructing two relevant states more than 2ε apart while its

259 region reads them identically is not resonant. Read forwards, a resonant
 260 boundary that reconstructs a pairwise separated family restricts injectively
 261 on it. Which states a region reads identically is fixed by where the region
 262 sits, before an encoding is chosen, so the resolution the family demands is
 263 the region's and no readout supplies it. The implication runs one way: a
 264 region that does resolve the family is not thereby shown to carry a resonant
 265 encoding, and E89 supplies no such resolution.

266 Resonance and resolution are jointly satisfiable without a local fold. Take
 267 the region to be the whole substrate and use the identity readout: restric-
 268 tion is the identity, so this boundary is resonant and reconstructs every de-
 269 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 270 E89's strict contraction premise on a nontrivial metric space. A properness
 271 condition on the region is therefore a separate requirement, and the fixed-
 272 contraction diameter bound remains in force when E89 is also required. With
 273 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 274 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 275 so the family is no larger than the δ -packing number of what the region car-
 276 ries. That is the shape a continuous family of states needs, and at the whole
 277 substrate it says nothing, because the region's range is everything. The gain
 278 L is declared and derived here for no physical readout, and at unbounded
 279 gain the bound says nothing.

280 5.3 Agreement after a change takes communication

281 Snapshot compatibility does not specify how quickly descriptions respond
282 when a shared quantity changes. In a synchronous network whose updates
283 read only the site’s own state and declared neighbours’ messages, two initial
284 configurations agreeing on a site’s causal past through T rounds produce the
285 same state there at round T . The deadline consequence is the one Section 2
286 states, and it follows from the reconstruction bound in Section 5.2: a report
287 from that site is a readout of a code that the two interventions leave identical.

288 A three-site delay line witnesses failure after one round and exact report-
289 ing after two; cutting the path prevents delivery at every round. A report
290 may still happen to be correct about one fixed world. The bound concerns
291 responsiveness across interventions.

292 Throughout this section *prediction* names a reconstruction map, and the
293 relevant family of states, the metric, the tolerance, the communication graph
294 and the deadline are all declared inputs, as are the represented variables and
295 the fidelity counted as accurate. Reading the map as temporal neural pre-
296 diction would require dynamics; reading the resulting code count as physical
297 memory capacity would require a hardware model, which a state-dependent
298 readout would have to extend.

299 6 What the conditions constrain in a digital 300 candidate

301 The results of Section 2 hold for any candidate, and for a large language
302 model (LLM) running on graphics processing units (GPUs) four of them are
303 checkable rather than merely stateable. Those four are where this framework
304 bears on claims made for such systems. It does not exclude them, and no re-
305 sult here could; what it does is fix what a claim of unity or self-representation
306 commits its maker to measuring.

307 **A quantitative bound for a linear bottleneck.** Let a finite spatial
308 coupling operator K have orthonormal eigenmodes e_i with $|\lambda_1| \geq \dots \geq |\lambda_N|$.
309 Every linear approximation A of rank at most d satisfies

$$\sum_{i=1}^N \|Ke_i - Ae_i\|^2 \geq \sum_{i=d+1}^N \lambda_i^2. \quad (11)$$

310 Retaining the first d modes attains equality. A linear encoder and readout
311 factoring through d real coordinates therefore pay this error. Applying the
312 bound requires a spectrum for the target and a rank constraint on the actual
313 approximation. Query/key dimension alone supplies no such constraint on
314 softmax attention: even a two-position rank-one logit matrix can yield rank-
315 two attention. The result prices an explicitly linear bottleneck in a finite
316 representation of a spatial kernel; it gives no universal gap between trans-

317 formers and continuous fields. Supplemental Material derives the bound and
318 checks the softmax counterexample.

319 A fixed synthetic task makes the scope visible. Two local views share
320 scene and self coordinates; the target is the global state and a self-relative
321 coordinate. On the parity-held-out family, a rank-two linear decoder’s input-
322 weighted predicted training-error floor is 2.000, and its held-out error is 2.000.
323 Equal numerical values do not make the training-law floor a held-out guaran-
324 tee. On a separate scene-family shift, the fitted full linear decoder has train-
325 ing error 0.000 and held-out error 2.667. The task’s declared target map
326 reconstructs the noiseless held-out inputs with error 0.000; the restricted
327 training family leaves one direction unidentified. A small causal attention
328 decoder and a coupled phase code also lose held-out accuracy on that shift
329 under their stated CPU fits; neither is governed by the linear rank floor. The
330 measured codes, overlap perturbations, self-state interventions and commu-
331 nication deadlines are described in Supplemental Material. These task results
332 rank no architecture family and supply no hardware-energy or experiential
333 claim.

334 **Codes, not reports.** Reconstruction across a declared family requires the
335 encoder to distinguish states more than 2ε apart when the target error is
336 ε (Eq. (10)). Verbal reports alone do not establish that condition. For
337 a proposed self-model, declare the family and fidelity, identify the internal
338 code under intervention, then test whether it separates the family. A token

sequence is one possible code; an activation region is another and requires its own measurement.

The same interface has an information ceiling. For any finite joint law of an internal state X and a length- k token sequence Y over vocabulary V , $I(X; Y) \leq H(Y) \leq k \log_2 |V|$ bits. The proof uses the full joint law and allows dependent emissions and zero-probability outcomes. A binary copy attains one bit per emitted binary symbol; an independent output conveys zero. This counts the declared output sequence, so a cache, timing signal or other accessible state must be included separately when it forms part of the channel.

The region, not the readout. If the code is a restriction of the candidate’s state to a region, that region must distinguish the relevant family before a readout is chosen. In one forward pass of a decoder-only transformer, the causal mask restricts position t to positions at or before t ; an intervention outside that causal past cannot be reconstructed from its activation alone. Across passes, prior output adds a separate channel. The bounded-fan-in communication result of Section 5 applies to a declared physical interconnect and deadline; a full-support attention graph does not satisfy its sparse-neighbourhood premise. Supplemental Material gives the graph bounds and their hardware scope.

Agreement about what. Unity requires the candidate’s local descriptions to agree on independently identified shared quantities (Section 4). Common

361 timing, a shared residual stream and fluent output supply none of that, for
362 the reason Section 2 gives: coordination bounds content disagreement only
363 through encoders already related on overlaps. Which quantities two parts
364 of a model describe in common is a decoding question, carrying the failure
365 modes of Section 8. The cover is a choice, and choosing it to secure agreement
366 empties the claim.

367 **Physical scope.** Digital representations can satisfy compatibility and re-
368 construction. The coupling-gated route additionally requires a calibrated
369 physical coupling, its noise regime and an independently identified relation
370 from phase order to shared content. An attention matrix or a shared clock
371 supplies none of these; the cortical proposal carries the same physical obliga-
372 tions. Finite-resolution capacity depends on code-space volume, resolution
373 and readout gain, so continuity alone favours no substrate. Supplemental
374 Material gives the capacity and hardware-support comparisons. Neither the
375 installed-energy condition nor those comparisons exclude a digital realization.

376 The conditions impose measurements on a digital candidate but settle
377 no question of machine experience. Supplemental Material places this com-
378 parison beside biological naturalism, substrate-dependent computation and
379 current proposals about LLM organization.

380 7 Thermodynamics and composition

381 The phase and representation results concern coupling, content disagreement
 382 and reconstruction error. A finite change in phase order has a probability-
 383 current cost bound under the conditions below. Thermodynamic heat re-
 384 quires a separate physical identification of the current and reservoir. The
 385 installed-energy condition of Section 3.2 is necessary for its declared actu-
 386 ator; whether it restricts a physical candidate depends on measured mode
 387 prices, coupling reduction and diffusion. No composition edge derives co-
 388 herence, compatibility or reconstruction from a heat bound. Finite-register
 389 capacity obeys $H(p) \leq \log |X|$; erasure heat depends on specified system-
 390 bath assumptions (Landauer, 1961; Norton, 2005; Shenker, 2000). A separate
 391 boundary result concerns a field reaching disconnected vacuum components
 392 (Kibble, 1976). Table 1 states the connecting assumptions.

393 7.1 Changing order carries a probability-current cost

394 For a smooth positive normalized periodic phase density ρ satisfying $\partial_t \rho =$
 395 $-\partial_\theta J$, with diffusion $D > 0$, define $\sigma_J = \int_{-\pi}^{\pi} J^2 / (D\rho) d\theta$ and $r = |\int e^{i\theta} \rho d\theta|$.
 396 Differentiation under the integral gives $\dot{r}^2 \leq D\sigma_J(1-r^2)$ wherever $0 < r < 1$.
 397 A smooth path staying in that interval therefore obeys

$$\int_0^\tau \sigma_J dt \geq \frac{[\arcsin r(\tau) - \arcsin r(0)]^2}{D\tau}. \quad (12)$$

Endpoint order, duration, diffusion and the probability current determine this floor. In an exploratory finite-volume onset calculation, a threshold chosen after inspecting pilot runs gives current cost 0.120 versus a floor of 0.046. These numbers describe the discretized model, not thermal or metabolic power.

The current $J = v\rho - D\partial_\theta\rho$ also satisfies

$$\left(\int_{-\pi}^{\pi} J d\theta\right)^2 \leq D\sigma_J. \quad (13)$$

A uniform density under constant drive ω attains $\sigma_J = \omega^2/D$. The identical-frequency gradient model has positive self-consistent stationary order above $K = 2D$ with $J = 0$ and $\sigma_J = 0$. Stationary order therefore has no positive current-cost floor. Nor does the bound price an externally chosen $\dot{K}$: in the identical-frequency zero-drive model, the uniform density remains an exact zero-current solution under any coupling schedule. A cortical power claim needs the physical current, reservoir and thermal conversion to be identified. This current functional is distinct from the substrate squared-drift objective used in the plasticity controls.

7.2 Feedback changes the information constraint

For passive memory X_t driven by an environmental signal S_t , the predictive bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (14)$$

416 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 417 et al., 2012). Zero retained information also satisfies it, so a task and a
 418 learning rule require additional specification. Relating this bound to Lan-
 419 dauer heat requires a common thermal scale and budget, and memory waste
 420 bounded by erased entropy.

421 An acting agent changes the signal it will receive. Specify policy, world,
 422 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{15}$$

423 The second conditional term vanishes in the passive model. With feedback,
 424 the signed difference can be negative as action creates correlations with the
 425 future. A causal reading requires an environmental state sufficient to describe
 426 the relevant dynamics.

427 For a controlled substep, the active thermodynamic bound instead uses
 428 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 429 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with thermal scale
 430 $k_B T > 0$ and heat positive into the reservoir (Horowitz and Esposito, 2014;
 431 Ito and Sagawa, 2013). Every forward path of positive probability must have
 432 a positive-probability reverse for this finite identity. A missing reverse gives
 433 infinite divergence. The supplement develops the support conditions and
 434 composes actuation with memory updates on their shared intermediate law.

435 **7.3 Learning and repeated operation need explicit re-** 436 **sources**

437 The finite models in Supplemental Material separate policy selection, learning
438 from a phase-derived sensor and repeated operation. A binary closed-loop
439 witness lets a register install the phase energy landscape, lets the phase evolve
440 under a bath-supported channel, senses the new phase and updates that same
441 register. A fixed decoder trained on one family predicts the next phase on
442 a held-out family with accuracy 0.783 under an informative sensor, versus
443 0.552 under a matched blind sensor. Their marginal phase order is the same.
444 The executed law also charges installation, sensing, reset and preparation
445 against a finite work store; phase-flip bath heat is recorded separately. The
446 symbol task and mode prices are supplied toy inputs, so this does not identify
447 a cortical mechanism or a heat cost for the full sensing and reset protocol.

448 For a complete cycle with a common energy observable $\mathcal{E}$ and thermal
449 scale $k_B T$, positive forward and reverse path support, and local detailed
450 balance at each stage, the conditional joint-law budget is

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (16)$$

451 when expected cumulative work stays within the supplied allowance B at
452 every prefix. This bounds expectations. The numerical witness specifies a
453 bath for the phase transition; its sensing, reset and preparation prices do
454 not establish reverse protocols or heat for the complete cycle. Separate con-

455 instructions track a store on each trajectory and specify when a mean allowance
 456 represents an available resource; an expected budget alone permits overdrawn
 457 paths.

458 For the declared process, a finite source with positive net cost funds
 459 finitely many cycles. A common physical ledger for the sensing and reset
 460 channels, and a cortical calibration of every mode price, remain separate
 461 obligations.

462 **7.4 What the composition establishes**

463 The passive and feedback composition theorems each accept eight connecting
 464 implications (Table 1). In the active branch, a named substep at the register’s
 465 thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (17)$$

466 with B the register’s heat. The upper allocation follows from a ledger whose
 467 operations deliver that total heat to one reservoir and whose other opera-
 468 tions do not increase joint entropy. Those ledger assumptions are physical
 469 inputs. Neither Landauer’s lower bound nor microscopic reversibility sup-
 470 plies them. Finite two-bit witnesses exhibit positive forward and reverse
 471 elementary paths, local detailed balance at one thermal scale, and interac-
 472 tion energies of order $k_B T$; these conditions alone do not quantify proximity
 473 to equilibrium. Cortex has not been shown to satisfy these premises. The

474 illustrative cortical metabolic expenditure over one region and one phase-
 475 diffusion time is 2.4×10^{14} times kT at body temperature, but identifying it
 476 with the register heat budget B requires a common physical ledger. Read-
 477 ing the register at the scale where its premises are physical, such as a single
 478 channel, requires E45Active to carry that step’s budget to the convergence
 479 of a continuum kernel. The active branch is an alternative route to the same
 480 node rather than a stronger one, and no result of Section 3.2 passes through
 481 it: the installed-energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.

Table 1 (continued)

Connection	Status	Required assumption
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .

Table 1 (continued)

Connection	Status	Required assumption
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and read-out. They conclude that the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named

497 in one place and machine-checked together, and not that coherence follows
498 from installed energy or self-representation from coherence.

499 Joint witnesses combine a register, boundary, refining mesh and small
500 spatial cover; active witnesses add controlled thermal updates and a priced
501 installation. They establish satisfiability under their stated identifications.
502 Mesh convergence follows from refinement data, and the witnesses establish
503 no common cortical mechanism. Supplemental Material maps the claims to
504 their Lean declarations and records the counterexamples. The development
505 declares no additional axioms beyond its libraries' foundational axioms.

506 **8 What the observations establish**

507 The composition holds under its premises. Whether observation can estab-
508 lish those premises is a separate question, and the three requirements it
509 connects concern different objects: phase dynamics, local descriptions and
510 internal encodings. Their observables must retain these distinctions. The
511 following controls show which mechanisms remain indistinguishable, which
512 temporal approximations fail, which states one observable cannot separate,
513 and what independent content measurements must establish. Each is a finite
514 numerical regime on a declared model, with full parameters and controls in
515 Supplemental Material, and none of them calibrates a cortical quantity.

516 8.1 Stationary distributions leave the mechanism un- 517 determined

518 Equation (2) gives a concentration–coherence relation with no fitted shape
519 parameter. Estimate r from circular resultant length and a from density
520 shape: inverting R from the measured r would make agreement automatic.
521 Even independently estimated agreement identifies a statistical family shared
522 by different mechanisms.

523 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
524 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
525 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
526 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
527 and diffusion D_0 have that same density. Their stationary distributions co-
528 incide window by window despite different causes.

529 Resolved dynamics supply additional information. In the ideal model
530 with continuous latent paths, quadratic variation accumulates at rate $2D$
531 and conditional increment means constrain drift; a filtered observation of
532 the phase needs its own observation model before either is read off. Separat-
533 ing endogenous coupling from a matched common drive additionally requires
534 a measured drive or controlled perturbation; identifying electromagnetic cou-
535 pling requires evidence specific to that interaction. Supplemental Material
536 gives the equivalence construction and its numerical check.

537 8.2 Finite ramps change the observable onset

538 The square-root law in Eq. (3) describes stationary solutions. A changing
539 coupling requires time for the phase distribution to approach those solutions.
540 In a deterministic mean-field control that crosses the threshold symmetrically,
541 none of the scanned rates meets the declared tracking tolerance: slowing
542 the crossing does not remove the terminal residual under that initialization.
543 A slow control variable therefore supplies no general guarantee of branch
544 tracking at onset. Supplemental Material gives the residual, rate range and
545 admissible rates away from threshold.

546 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
547 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
548 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
549 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
550 6 uncensored legs supply the delay scaling below; the decade-spaced ones
551 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
552 onset exponents differ from those of the exact stationary branch evaluated
553 on the same grids. At the fastest speed the ensemble mean never reaches the
554 fit window, so its fitted slope does not estimate stationary onset.

555 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
556 makes the apparent exponent depend on ramp speed and fitting window.
557 The specified deviation-based critical ramp speed is not bracketed by this
558 sweep. With fluctuating coupling, the relevant stationary comparison also
559 depends on rate: the mean coupling approximates order in the tested fast

560 regime, while averaging stationary order over coupling works for the tested
 561 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 562 neither approximation.

563 **The prediction.** Over the 6 uncensored legs a finite-range fit gives $\Delta K \propto$
 564 $v^{0.444}$. The interval from fitting those means is $[0.394, 0.494]$; resampling
 565 replicas independently within each speed gives a 95% interval $[0.398, 0.496]$,
 566 with fraction 0.019 of 10000 bootstrap fits at or above $1/2$. This fraction
 567 describes the resampling distribution under the fitted protocol; it is not a
 568 test of the ideal exponent. A deterministic mean-field run seeded at the
 569 fluctuation floor gives exponent 0.425 with the same estimator and escape
 570 criterion; tightening its criterion to $r \geq 0.05$ gives 0.490. These controls
 571 show that the criterion affects the fitted exponent, but do not prove that it
 572 accounts for the entire finite-population shortfall. The larger-population low-
 573 threshold fit has substantial log residuals; matched-speed fits do not converge
 574 toward one-half across the tested sizes and criteria. Some trajectories also
 575 cross the absolute escape level before K_c , so the first postcritical crossing can
 576 yield a zero or grid-limited delay. This finite sweep establishes no common
 577 asymptotic exponent.

578 The ideal delayed-onset calculation motivates a square-root scaling hy-
 579 pothesis; its exponent is generic to suitable bifurcations and does not iden-
 580 tify the coupling medium. A within-subject emergence design can vary the
 581 propofol taper and test how an operational onset changes with that manipu-

lation, but interpreting the result as a coupling-rate law requires a measured
or calibrated relation between taper and $\dot{K}$. The escape criterion must be
set relative to each recording's pre-threshold fluctuation scale. Section 8.1
describes how D could be estimated from resolved phase traces.

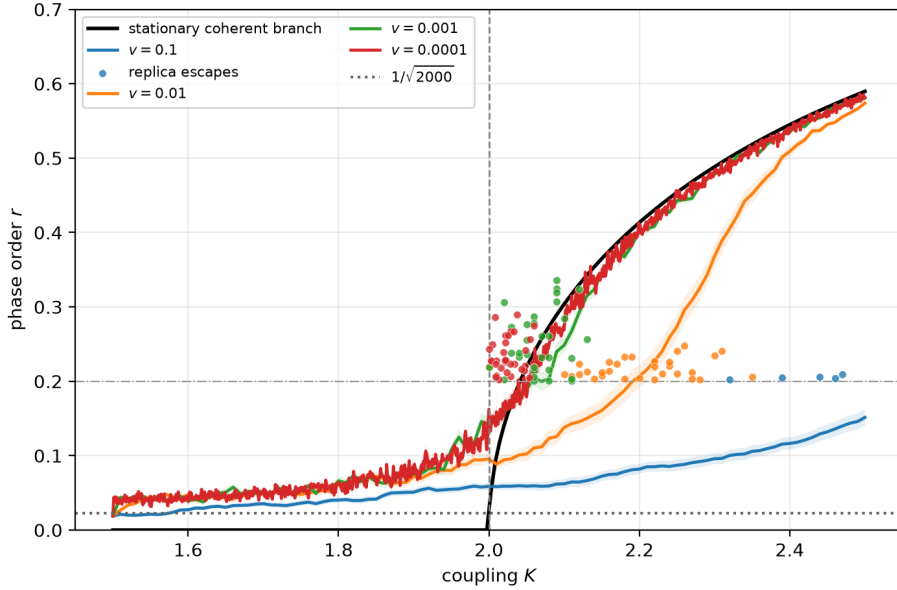


Figure 3: Finite-rate departures from the stationary branch. Curves show ensemble-mean order for four ramp speeds v , with one-standard-error bands. Coloured points mark each replica's first sustained escape, plotted at its recorded order; censored replicas have no point. The dashed vertical line is the static threshold, the dash-dotted horizontal line is the escape criterion, and the dotted horizontal line is the finite-size reference. The replica-bootstrap interval for the delay exponent is reported in the text.

Additional controls address integration bias, population size and spatial
or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 $[1.958, 2.009]$ contains 2.

589 8.3 A winding state leaves the order parameter am- 590 biguous

591 The global order parameter r discards spatial arrangement. On a periodic
592 sheet, a phase field can wind once around the sheet and have $r = 0$ while
593 neighbouring sites remain locked. At identical frequencies, every such wind-
594 ing is stationary for an isotropic symmetric kernel: contributions at opposite
595 separations cancel. It violates the quarter-turn initial-spread premise of the
596 convergence result, which is why that result can still select the uniform state.

597 In the spatial control, the winding persists below the interpolated de-
598 cay length 0.1871 mm and unwinds above it (Fig. 4). At 0.1 mm, uniform
599 and winding starts under the same coupling, diffusion and kernel end at
600 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
601 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
602 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
603 realizations. These are finite controls, not a cortical wave identification or a
604 derived transition.

605 Linearization about a winding reweights each separation's coupling by
606 the cosine of its phase advance. The resulting symmetric kernel has explicit
607 mode rates: nonnegative reweighted coupling gives no growing mode, while
608 a positive rate destabilizes that winding. The criterion concerns linear rates,
609 not asymptotic stability, and the measured retention boundary need not
610 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$

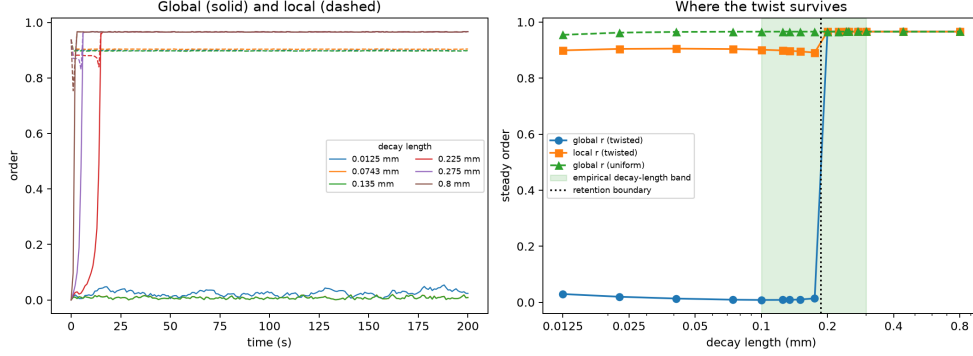


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

611 0, where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 612 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 613 winding from that model even though the phase-only equations retain it as
 614 a stationary state.

615 Neither model turns these states into measured cortical spirals. The pe-
 616 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
 617 no propagation speed, and the amplitude variable has no calibrated relation
 618 to a recorded envelope. Supplemental Material gives the spectral, amplitude
 619 and defect results. Locally, Eq. (6) can still bound content disagreement on a
 620 winding patch when shared encoders are supplied, although its global $r = 0$

form is uninformative. A nonzero ring winding number q also obstructs a global real-valued phase lift; it has no identified content meaning here. Thus an onset criterion based only on r can register a change of basin as well as a change of coupling. The protocol of Section 9.3 needs a spatial phase observable.

8.4 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might supply representational structure, but its objective must reward the relevant agreement. For symmetric coupling, pairwise sine terms cancel in the population mean drift. Minimizing squared drift at that fixed mean therefore rewards equalizing drift across nodes. In the frequency-cluster control, this favors coupling between unlike-frequency clusters, whereas the diagnostic environmental template rewards coupling within clusters.

The formalization proves descent of the squared-drift objective for a specified gradient flow at fixed phases. Its functional differs from thermodynamic entropy production: at a coherent stationary density the probability current vanishes while phase-averaged squared drift is positive. Supplemental Material gives the operator, its gradient and the conditions of the descent result.

The finite control co-evolves noisy phases and symmetric, nonnegative couplings at fixed total resource. Its template is a diagnostic covariance

of latent signals, never a training input. Across three illustrative seeds, second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel controls are reported in Supplemental Material.

The immediate post-update objective reduction disappears when averaged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on synthetic data separates these

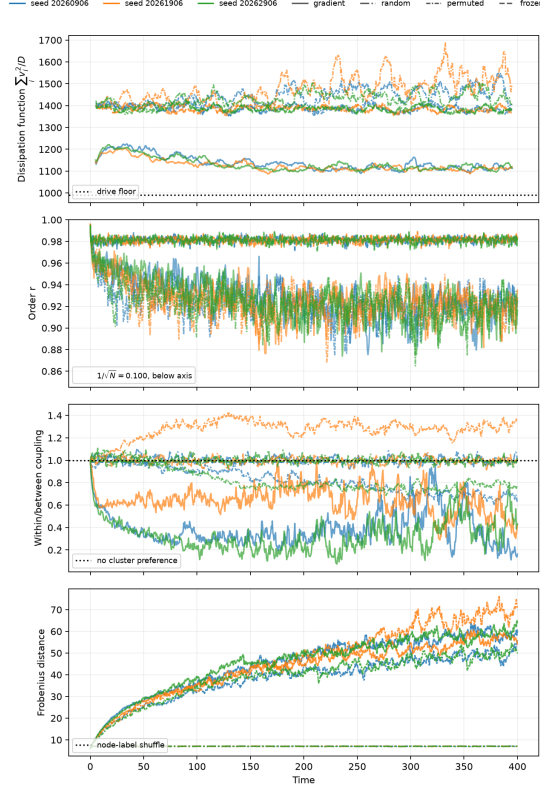


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

664 cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial;
665 the compatible construction scores 0.476 on an interval containing one half.
666 Shared-prior shrinkage can nevertheless manufacture agreement, and regional
667 bias can manufacture disagreement. Per-patch decoding diagnostics detect
668 these failures; the compatibility statistic alone does not.

669 Phase agreement cannot replace that measurement. On constructed con-
670 tent, phase carries a share 1.000 of the content entropy when content is built
671 from it and 0.000 when it is not. A phase-derived statistic reports com-
672 patibility on phase-locked patches whose local descriptions disagree. The
673 implemented content statistic detects their incompatibility. These controls
674 validate a measurement on known constructions; a neural content model and
675 decoder remain to be established.

676 Choosing the cover also chooses a candidate subject: trivial descriptions
677 can agree while carrying little content, and a partition can conceal disagree-
678 ment. The useful choice must therefore be constrained by what neural ac-
679 tivity encodes. The exact composition assumes compatibility of its supplied
680 cover. The content mechanisms above identify additional hypotheses under
681 which agreement can arise; they leave the empirical choice of content and
682 the connection to that cover explicit.

683 **8.6 Estimator calibration on exploratory EEG**

684 The concentration range and the dependence structure decide whether the
685 concentration-coherence relation can be tested at all. Under the specified

log-density estimator and bias-floor criterion, discrimination at $a = 0.542$ requires 3000 independent sites in the scanned designs, rising to 31000 when sites share a mean in groups of 10. These counts describe that estimator and dependence model at a declared design concentration; the intended recording requires its own calibration. Supplemental Material reports the bin-count controls and the full dependence-aware design.

The estimators were also exercised on public propofol-sedation EEG (Bajwa et al., 2025): 61 adjacent bipolar pairs formed from 62 scalp channels, band-passed, Hilbert-transformed and pooled over pairs and 100-ms windows. Across 8 subjects the pooled concentration spans 0.087–0.142, and the residual from the Bessel relation is small. That agreement is uninformative. Pooling over a window does not remove the common rotation of the phases, so a perfectly phase-locked 10-Hz signal passed through the same pipeline returns $r = 0.0001$ and $a = 0.0004$ with zero residual, although its instantaneous spatial resultant is one. The bipolar montage also removes spatial coherence: a field whose raw channels have instantaneous resultant 0.89 gives 0.06 across bipolar pairs and 0.007 pooled. Any nearly uniform pooled distribution satisfies $r \approx a/2$, and over the observed range the Bessel ratio departs from $a/2$ by at most 0.0002. The exercise therefore checks the estimator’s behaviour and establishes no spatial concentration, coherence level or test of the stationary density. Sedation recordings also leave the awakening hypothesis untested.

An informative dataset therefore needs an estimand that removes the

709 common rotation, a montage whose effect on spatial phase is modelled, suf-
710 ficient concentration range or precision, calibrated estimators and an ob-
711 servation model for dependence and source mixing. Intracranial recordings
712 (ECoG, sEEG or LFP) or narrowband burst analysis may reach a more
713 informative concentration range, but that range must be measured. The
714 design counts above are calculations for the specified estimator, not sam-
715 ple requirements transferable to another recording. Public mouse calcium
716 imaging provides thousands of spatially located neurons across waking, sleep
717 and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the
718 phase-estimation band used here. Site count, state coverage and bandwidth
719 are separate requirements.

720 **9 The cortical hypothesis and an awakening**

721 **test**

722 The cortical proposal identifies a physical source of coupling and asks whether
723 its changes help explain recovery. The tests in Section 8 determine what
724 observations must distinguish before a phase trace can support this identifi-
725 cation.

726 9.1 Physical identification and calibration

727 The cortical hypothesis identifies part of effective coupling with endogenous
728 extracellular electromagnetic interactions. Fields can modulate spike timing
729 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
730 propose coordination of distributed neural information (Pinotsis & Miller,
731 2022; McFadden, 2020). Phase organization under propofol motivates study-
732 ing this regime (Bastos et al., 2021), but does not measure coupling.

733 The formal field predicate requires a jointly continuous kernel, a normal-
734 ized substrate, coupling equal to the kernel average, and no single site pro-
735 viding more than half that strength. Toy constructions satisfy it. A cortical
736 realization must specify the scale at which the kernel and its reduction ap-
737 ply: calcium imaging reports different state-dependent modularity patterns
738 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

739 Cortical columns are the obvious candidate unit, and they answer one
740 of the two demands the formalism makes. As the oscillator index they are
741 a serious proposal: they are delineated by response properties and anatomy
742 rather than by the agreement they would be used to establish (Mountcastle,
743 1997), and a phase can be estimated from a column-sized territory at the
744 electrode spacings used for cortical wave measurements (Townsend et al.,
745 2015). As the content cover they are a poor one, for three separate reasons.
746 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
747 the compatibility obligation without constraining anything (Section 4). The
748 cover condition additionally requires the patches to exhaust the substrate,

749 while much of association cortex has no accepted columnar parcellation. And
750 the column's standing as a canonical cortical unit is contested (Horton &
751 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
752 supplies the oscillator index; the content cover remains an open empirical
753 question, which the bridge assumption E78 records.

754 The spatial reduction is also testable. A row-sum aggregation at zero
755 diagonal tracks the uniform-coupling control over the tested spatial decay
756 range, while frequency heterogeneity displaces the operational threshold. The
757 fit window biases both estimates relative to $2D$; agreement with the control
758 is therefore the supported conclusion. Supplemental Material reports the
759 reduction, its numerical error and the distinct assumptions needed for the
760 kernel-average identification in Eq. (5).

761 The field commitment requires physical evidence beyond a choice of math-
762 ematical representation, and the available support comparisons do not supply
763 it. A finite matrix spread over positive-mass cells preserves its total cou-
764 pling and phase correlation exactly, while point-supported vanishing under
765 an atomless measure and a disadvantage from excluding a better-correlated
766 pair concern specified support constraints; Section 6 reads the same compar-
767 isons against a digital candidate. Neither supplies energy prices.

768 A spike-time shift alone supplies no coupling rate. The dimensionless
769 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and

770 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (18)$$

771 This rate describes how induced timing shifts accumulate into drift. Nei-
772 ther it nor the installation factor κ follows from a heat bound. Testing the
773 physical threshold requires independent estimates of the interaction kernel,
774 its aggregation and diffusion in the same cortical regime.

775 **9.2 Extracellular geometry and recovery**

776 The proposed control variable is extracellular geometry. Extracellular vol-
777 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
778 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
779 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
780 at $0.3\text{--}0.6 \text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
781 an effective-medium description predicts that a larger extracellular fraction
782 raises conductivity and reduces field amplitude.

783 The coupling-gated hypothesis consequently predicts that extracellular
784 contraction on awakening raises E and, holding the other factors in Eq. (18)
785 fixed, raises effective coupling. This argument supplies a direction; the
786 geometry–coupling relation, its magnitude and its role in human recovery
787 require measurement.

788 For a smooth crossing at nonzero speed, evaluating the stationary branch

789 along the changing control gives a deterministic tracking approximation that
 790 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (19)$$

791 This is a deterministic tracking approximation from Eq. (3) at fixed D ; the
 792 crossing controls in Section 8.2 show that it fails as an observed-onset ap-
 793 proximation under the tested initialization. A separation of geometry and
 794 phase time scales can support tracking away from threshold. The awakening
 795 hypothesis needs a calibrated dynamical prediction for an operational onset
 796 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
 797 recovery latency.

798 **9.3 A protocol for the awakening experiment**

799 Neural inertia describes a behavioural barrier and hysteresis between anaes-
 800 thetic induction and emergence (Friedman et al., 2010); recovery can pass
 801 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
 802 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
 803 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
 804 mechanism from a synaptic one. A possible discriminator is spatial onset
 805 relative to independently estimated local field coupling and network connec-
 806 tivity. The present mean-field model has no spatially varying κ and derives
 807 no nucleation order, so a field-based onset map needs a separate model spec-

808 ified before the observations are inspected.

809 For that comparison, partition the recording into prespecified regions i
810 and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
811 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
812 and a connectivity score c_i from a separately estimated structural network.
813 For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
814 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic
815 approximation predicts

$$\hat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (20)$$

816 Predicted onset is the first crossing of a region-specific threshold fixed from
817 pre-onset fluctuations. Fit the two scales on training sessions, then com-
818 pare held-out onset times and site order under the same observation model.
819 Larger field scores predict earlier onset only when baseline order and diffu-
820 sion do not reverse the ranking. This declared approximation omits interre-
821 gional coupling and saturation; a successful ranking would support its field
822 covariate, not prove ephaptic causation. Supplemental Material specifies the
823 controls and failure conditions.

824 A within-subject protocol can manipulate emergence rate, with spatially
825 resolved recordings to compare the proposed field and connectivity models:

- 826 1. **Manipulate emergence rate.** Vary the propofol infusion taper
827 across sessions within the same subject, recording drug concentrations

828 or a validated pharmacokinetic proxy to quantify the manipulation.

829 **2. Measure behavioural recovery.** Record time-to-response to a stan-
830 dard stimulus as the independent behavioural endpoint, rather than
831 estimating ΔK directly.

832 **3. Resolve spatial onset.** Identify anatomical locations where coher-
833 ence first crosses the baseline-scaled threshold and score the two maps
834 of Eq. (20) on held-out sessions. The two covariate maps are fixed
835 before the outcome sessions are inspected.

836 **4. Match physical controls.** Confirm that the observed coherence-to-
837 amplitude relation holds under the specific local diffusion D and ob-
838 servation noise. The model’s quadratic-variation rate $2D$ applies to
839 the latent phase; a band-limited analytic phase is smooth below its
840 correlation time, and on a constructed field its squared-increment rate
841 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
842 fore requires an explicit observation model for filtering and sampling.

843 Predeclare the recording window, regional fluctuation thresholds, and a
844 shared timing-error model. Fit the two scales and that observation model
845 on training sessions only; on held-out sessions, score the observed site-onset
846 times and sites that never cross by the same regional predictive rule. Report
847 site order as a secondary outcome and behavioural time-to-response sepa-
848 rately. Sessions without a crossing are censored observations, not zero delays
849 or exclusions. Compare the two scores only where independently estimated

850 field and connectivity maps predict distinguishable onset patterns at the
851 recording resolution. A held-out field-model advantage would support that
852 specified spatial predictor; it would not identify an ephaptic cause, shared
853 content or experience. Supplemental Material gives the controls and pilot
854 quantities needed to judge whether the comparison is informative.

855 **10 Discussion**

856 **10.1 Unity and a minimal self as interpretive motiva-** 857 **tion**

858 We propose the state glued from compatible local descriptions as a formal
859 correlate of conscious unity, and its accurate internal reconstruction as a cor-
860 relate of a minimal self in the sense of a pre-reflective point of view on the
861 scene (Gallagher, 2000). In the cup example, this would mean one description
862 that preserves the cup’s relation to the perceiver as well as the scene. Com-
863 patibility permits distinct contents and uncertainty; the shared quantities
864 must be chosen from independent evidence about what each region encodes.
865 The reconstruction need not be a verbal or autobiographical account. Its
866 proposed relation to experienced perspective remains an interpretation. Nei-
867 ther identification is among the eight hypotheses in Table 1, and satisfying
868 them would not measure experience.

869 **10.2 A comparison that could disconfirm the link to** 870 **content**

871 The framework predicts that conscious content tracks compatibility, not co-
872 herence. Present matched stimuli that are reported on some trials and not
873 on others, for example at masking or rivalry threshold, and fit decoders for a
874 shared stimulus quantity independently on two overlapping cortical regions.
875 Compare decoded agreement on the overlap between reported and unreported
876 trials after matching their phase coherence. The proposed link predicts higher
877 agreement on reported trials at equal coherence. Equal agreement, or agree-
878 ment that follows coherence and not report, counts against it. A self-model
879 comparison has the same form: reconstruction of perceiver-relative variables,
880 such as hand position relative to the target, from a declared local encoding
881 should be more accurate on reported trials. These are correlational tests of
882 the proposed correlates; report is itself an access measure, and a no-report
883 design would be needed to separate content from access.

884 **10.3 What a test of the full proposal must establish**

885 A test needs separate estimates of effective coupling and diffusion, shared
886 content on overlapping regions, and reconstruction from a specified encoding
887 at a declared fidelity. The cover, relevant state family and decoder must
888 be fixed without selecting them for agreement. An independent experiential
889 endpoint is needed to evaluate the interpretation; absence of a report alone

890 does not define absence of experience. A unified episode with incompati-
891 ble decoded descriptions would challenge the unity identification. A unified
892 episode lacking the proposed self-encoding would challenge the minimal-self
893 claim. Conversely, satisfying the mathematical conditions without such an
894 episode would challenge their sufficiency as correlates.

895 For the physical route, cortical actuator modes and prices, the geometry–
896 coupling relation and a dynamical onset model remain to be measured. The
897 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
898 diction. Controlled changes in coordination and independently decoded con-
899 tent agreement would address the representational connection. The synthetic
900 decoder exercise calibrates part of these measurements, and the EEG exercise
901 identifies what a pooled bipolar estimand cannot measure; neither identifies
902 a neural content code or tests experience.

903 **10.4 Explanatory scope**

904 Integrated-information, global-workspace and predictive-processing accounts
905 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
906 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
907 relations among coupling, local content and reconstruction. Its formal field
908 condition requires a continuous kernel, a normalized substrate, a specified
909 average coupling and noise, and a site-contribution bound. Those conditions
910 do not identify an electromagnetic substrate; the extracellular interpretation
911 needs the separate calibration of Section 9. The joint witness shows that

the formal hypotheses can hold together in a declared model. A cortical realization and an experiential interpretation require additional evidence.

Three limitations bound every claim above. Synchrony is not itself an experiential measure, and nothing here establishes a consciousness criterion: the conditions are proposed correlates to be tested against report or another independent endpoint. The present EEG analysis checks a measurement pipeline and does not validate any experiential claim. The formal results are conditional on declared encoders, covers and maps whose cortical counterparts are unmeasured.

The remaining explanatory question is whether these relations occur in neural activity and matter for experience. The framework identifies candidate correlates and tests of their physical and representational premises. It does not explain why satisfying them would be accompanied by experience.

Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	§2, Supp.	Phase diffusion in inverse time; phase spread $D \leq \pi/2$

Table 2 (continued)

Symbol	Defined	Meaning
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§3.2, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Section 6	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (17)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (18)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops

Table 2 (continued)

Symbol	Defined	Meaning
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
$\mathcal{E}$	Eq. (16)	Energy observable shared by the stages of a continuing protocol
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (18)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

926 Data and code availability

927 The Lean development, simulation scripts, tests and saved numerical
928 summaries are maintained in the GitHub repository `bobaseb/coupling_`
929 `gated_cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and
930 Mathlib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publi-
931 cation simulation values are generated from saved summaries without re-
932 running production sweeps. The exploratory EEG analysis uses Open-
933 Neuro dataset `ds005620` (Bajwa et al., 2025) (subjects 1010, 1016, 1022,
934 1033, 1045, 1056, resting awake and sedation runs 1–2); preprocessing
935 and selection are described in Supplemental Material and implemented in

936 `simulations/empirical_collapse.py`. The repository contains the repro-
937 ducible analysis and does not redistribute the raw recordings.

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941 **Use of AI tools**

942 AI assistance to this work was substantial and is not separable by task. Sev-
943 eral coding assistants and chat interfaces were used in an interleaved way
944 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
945 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
946 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
947 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
948 accessed with the Hermes CLI. Between them they contributed to the Lean
949 development, the simulation and analysis code, and the drafting and revision
950 of both documents. Because the tools were used in combination and in alter-
951 nation, no section, proof, figure or result is attributable to any one of them,
952 and none is claimed to be. The author is responsible for all content, including
953 everything a tool produced. Machine checking establishes the stated Lean
954 results under their hypotheses; it does not validate biological interpretations

955 or replace scientific review of generated code and prose.

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